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About the Project

BawarchiKhana (Urdu: باورچی خانہ — literally "kitchen") is an intelligent cooking assistant platform built specifically for Pakistan and the broader South Asian market. It bridges the gap between everyday Pakistani cooking and modern AI technology.

The platform helps users:

- Discover **halal recipes** tailored to ingredients they already have
 - Reduce **food waste** through intelligent pantry management
 - Get **interactive step-by-step cooking guidance** with Urdu voice support
 - Plan **weekly meals** powered by Google Gemini AI
 - Export **meal plans and recipes** to Google Sheets and Google Drive
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Features

AI-Powered Capabilities

Feature	Description
Recipe Suggestions	Get recipe ideas based on your pantry ingredients
Meal Planning	AI-generated weekly meal plans tailored to preferences
AI Vision	Upload food photos to identify ingredients and get recipes
Cooking Copilot	Interactive step-by-step cooking guidance with AI
Ingredient Substitute	Instantly find halal substitutes for any ingredient

User Management

- OTP-based phone authentication via Twilio
- Google OAuth2 sign-in
- JWT-secured sessions with configurable expiry
- User profile and cooking history tracking
- Pantry management (add/update/remove ingredients)

Google Integrations

- Export weekly meal plans to Google Sheets
- Save recipes to Google Drive
- Import ingredient lists from Drive
- OAuth2-secured Google account linking

Payments (Premium)

- JazzCash payment gateway integration
- EasyPaisa payment gateway integration
- Webhook support for payment confirmation
- Freemium model with premium recipe unlock

Tech Stack

Backend

Technology	Version	Purpose
NestJS	v11	Main backend framework
TypeScript	v5.7	Language
Prisma ORM	v5.22	Database access layer
PostgreSQL	v18	Primary database
Google Gemini AI	v0.24	AI/ML capabilities
Passport.js	v0.7	Authentication middleware
JWT	v11	Token-based auth
ioredis	v5.10	Redis client (caching)
PDFKit	v0.18	PDF report generation
Twilio	SDK	OTP delivery

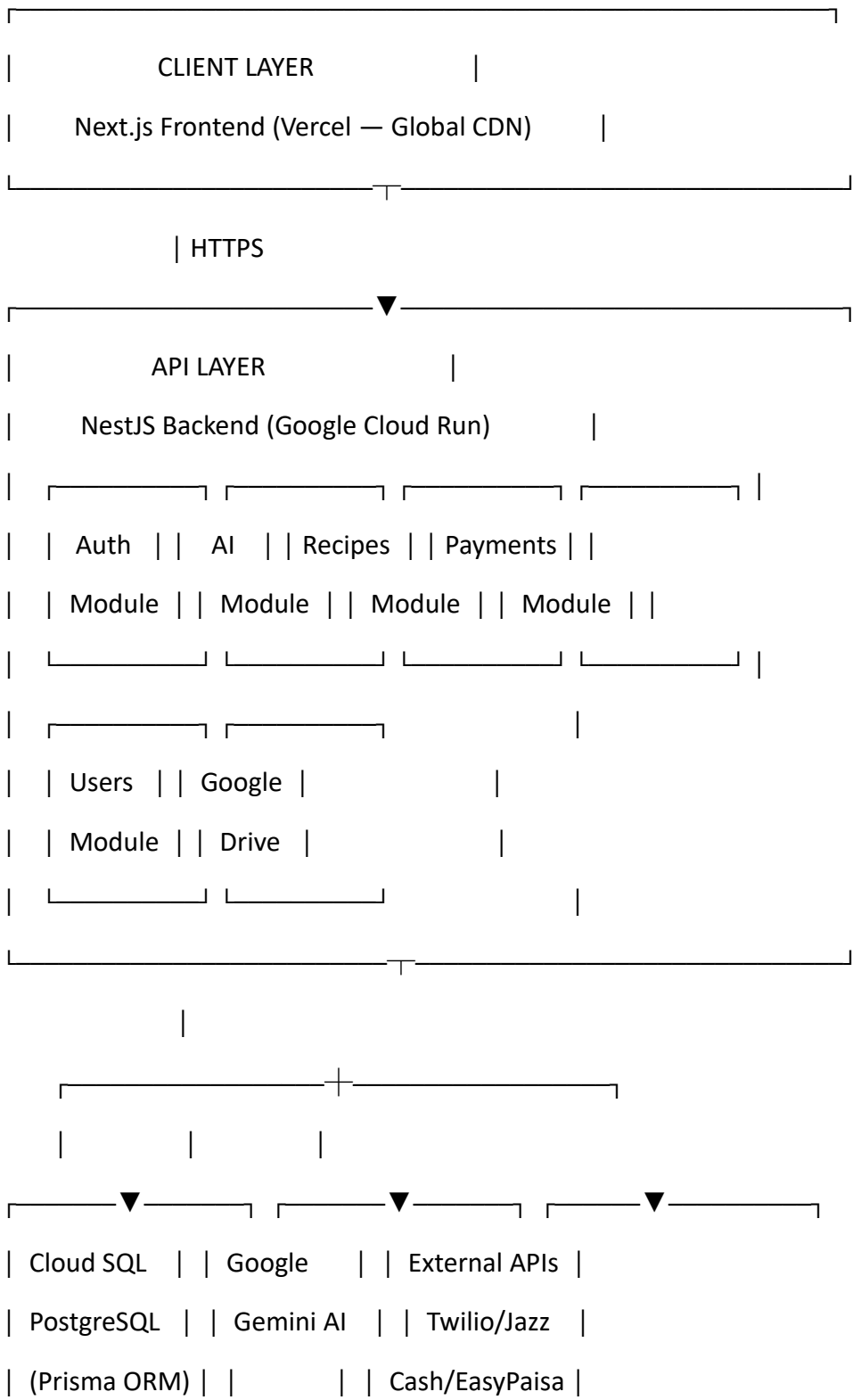
Frontend

Technology	Purpose
Next.js	React framework
TypeScript	Language
Vercel	Deployment & edge CDN

Infrastructure & DevOps

Technology	Purpose
Google Cloud Run	Serverless container hosting (asia-south1)
Google Cloud SQL	Managed PostgreSQL database
Google Artifact Registry	Docker image registry
Google Cloud Build	CI/CD pipeline
Docker	Containerization (node:22-alpine)

Architecture



API Endpoints Summary

Module	Endpoints
Auth	POST /auth/request-otp, POST /auth/verify-otp, GET /auth/google/callback
AI	POST /ai/suggest, POST /ai/meal-plan, POST /ai/vision, POST /ai/copilot, POST /ai/substitute
Recipes	POST /recipes/session, GET /recipes/session/:id, PATCH /recipes/session/:id/complete
Users	GET /users/profile, PATCH /users/pantry, GET /users/history
Payments	POST /payments/checkout, POST /payments/webhook
Google Drive	GET /google-drive/connect, GET /google-drive/files, POST /google-drive/export-recipe, POST /google-drive/export-weekly-plan, POST /google-drive/export-sheet, POST /google-drive/import

Getting Started

Prerequisites

- Node.js >= 22.x
- npm >= 10.x
- PostgreSQL database (local or Cloud SQL)
- Google Cloud project (for deployment)

Local Development

1. Clone the repository

```
git clone https://github.com/fuzailfaraz/BawarchiKhana.git
```

```
cd BawarchiKhana/backend
```

2. Install dependencies

```
npm install
```

3. Set up environment variables

```
cp .env.example .env
```

```
# Fill in your values (see Environment Variables section)
```

4. Run database migrations

```
npx prisma migrate dev
```

```
npx prisma generate
```

5. Start development server

```
npm run start:dev
```

The API will be available at <http://localhost:3001>

Frontend Setup

```
cd ../frontend
```

```
npm install
```

```
npm run dev
```

Frontend available at <http://localhost:3000>

Environment Variables

Create a .env file in the backend/ directory:

```
# Database
```

```
DATABASE_URL
```

```
# Redis
```

```
REDIS_URL
```

```
# Authentication
```

```
JWT_SECRET
```

```
JWT_EXPIRY
```

```
# AI
```

```
GeminiAI_API_KEY
```

JazzCash

EasyPaiza

Google OAuth

GOOGLE_CLIENT_ID

GOOGLE_CLIENT_SECRET

GOOGLE_REDIRECT_URI

App

FRONTEND_URL

PORT=3001

NODE_ENV=development

Deployment

Docker Build

cd backend

docker build -t bawarchikhana-backend:latest .

Deploy to Google Cloud Run

Build and push image:

gcloud builds submit --config cloudbuild.yaml --region=asia-south1 .

Deploy:

gcloud run deploy bawarchikhana-backend \

--image "asia-south1-docker.pkg.dev/PROJECT_ID/bawarchikhana-repo/bawarchikhana-backend:latest" \

--platform managed \

--region asia-south1 \

--allow-unauthenticated \

```
--add-cloudsql-instances PROJECT_ID:asia-south1:bawarchikhanadb \  
--set-env-vars "DATABASE_
```

Live URLs:

-  Frontend: <https://bawarchi-khana-pi.vercel.app>
-  Backend API: <https://bawarchikhana-backend-248619180591.asia-south1.run.app>

Database Schema

Key entities in the Prisma schema:

Users ———┐—— CookingSession
 ├—— PantryItem
 └—— Payment

Recipe ———┐—— CookingSession
 └—— Ingredient

MealPlan —┐—— MealPlanItem → Recipe

Roadmap

- [x] User authentication (OTP + Google OAuth)
- [x] AI recipe suggestions (Gemini)
- [x] AI meal planning
- [x] AI vision (food photo recognition)
- [x] AI cooking copilot
- [x] Ingredient substitutes
- [x] Cooking sessions
- [x] Pantry management

- ☒ Google Drive integration
 - ☒ Payment integration (JazzCash + EasyPaiza)
 - ☒ Cloud Run deployment
 - ☐ Urdu voice support (TTS)
 - ☐ Mobile app (React Native)
 - ☐ Community recipe sharing
 - ☐ Receipt scanning for pantry auto-update
 - ☐ Redis caching layer
 - ☐ Nutritional information per recipe
-

Contributing

1. Fork the repository
 2. Create your feature branch: `git checkout -b feature/AmazingFeature`
 3. Commit your changes: `git commit -m 'Add AmazingFeature'`
 4. Push to the branch: `git push origin feature/AmazingFeature`
 5. Open a Pull Request
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Contact

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Project Link: <https://github.com/fuzailfaraz/BawarchiKhana>